Glossary

Acute
An acute angle is an angle that is smaller than 90º.

Addition
Working out the sum of a group of numbers. Addition is represented by the + symbol, e.g. 2 + 3 = 5. The order the numbers are added in does not affect the answer: 2 + 3 = 3 + 2.

Adjacent
A term meaning "next to". In two-dimensional shapes two sides are adjacent if they are next to each other and meet at the same point (vertex). Two angles are adjacent if they share a vertex and a side.

Algebra
The use of letters or symbols in place of unknown numbers to generalize the relationship between them.

Alternate angle
Alternate angles are formed when two parallel lines are crossed by another straight line. They are the angles on the opposite sides of each of the lines. Alternate angles are equal.

Angle
The amount of turn between two lines that meet at a common point (the vertex). Angles are measured in degrees, for example, 45°.

Apex
The tip of something e.g. the vertex of a cone.

Arc
A curve that is part of the circumference of a circle.

Area
The amount of space within a two-dimensional outline. Area is measured in units squared, e.g. cm^2.

Arithmetic
Calculations involving addition, subtraction, multiplication, division, or combinations of these.

Average
The typical value of a group of numbers. There are three types of average: median, mode, and mean.

Axis (plural: axes)
Reference lines used in graphs to define coordinates and measure distances. The horizontal axis is the x-axis, the vertical axis is the y-axis.

Balance
Equality on every side, so that there is no unequal weighting, e.g. in an equation, the left-hand side of the equals sign must balance with the right-hand side.

Bar graph
A graph where quantities are represented by rectangles (bars), which are the same width but varying heights. A greater height means a greater amount.

Base
The base of a shape is its bottom edge. The base of a three-dimensional object is its bottom face.

Bearing
A compass reading. The angle measured clockwise from the North direction to the target direction, and given as 3 figures.

Bisect
To divide into two equal halves, e.g. to bisect an angle or a line.

Box-and-whisker diagram
A way to represent statistical data. The box is constructed from lines indicating where the lower quartile, median, and upper quartile measurements fall on a graph, and the whiskers mark the upper and lower limits of the range.

Brackets
1. Brackets indicate the order in which calculations must be done —calculations in brackets must be done first e.g. 2 x (4 +1) = 10.
2. Brackets mark a pair of numbers that are coordinates, e.g. (1, 1).
3. When a number appears before a bracketed calculation it means that the result of that calculation must be multiplied by that number.

Break even
In order to break even a business must earn as much money as its spends. At this point revenue and costs are equal.

Calculator
An electronic tool used to solve arithmetic.

Chart
An easy-to-read visual representation of data, such as a graph, table, or map.

Chord
A line that connects two different points on a curve, often on the circumference of a circle.

Circle
A round shape with only one edge, which is a constant distance from the centre point.

Circle graph
A circular graph in which segments represent different quantities.

Circumference
The edge of a circle.

Clockwise
A direction the same as that of a clock's hand.

Coefficient
The number in front of a letter in algebra. In the equation $x^2 + 5x + 6 = 0$ the coefficient of 5x is 5.

Common factor
A common factor of two or more numbers divides exactly into each of those numbers, e.g. 3 is a common factor of 6 and 18.

Compass
1. A magnetic instrument that shows the position of North and allows bearings to be found.
2. A tool that holds a pencil in a fixed position, allowing circles and arcs to be drawn.

Composite number
A number with more than two factors. A number is composite if it is not a prime number e.g. 4 is a composite factor as it has 1, 2, and 4 as factors.

Concave
Something curving inwards. A polygon is concave if one of its interior angles is greater than 180°.

Cone
A three-dimensional object with a circular base and a single point at its top.

Congruent/congruence
Two shapes are congruent if they are both the same shape and size.

Constant
A quantity that does not change and so has a fixed value, e.g. in the equation y = x + 2, the number 2 is a constant.